

Report from Program Committee QIP 2021

B.M. Terhal

Program Committee

35 m/w (same as last year)

Ryan Babbush | Google Research

Alexander Belov | University of Latvia

Sergey Bravyi | IBM's Watson Research Center

Anne Broadbent | University of Ottawa

Earl Campbell | Amazon & University of Sheffield

Roger Colbeck | University of York

Claude Crepeau | McGill University

Frederic Dupuis | University of Montreal

Bill Fefferman | University of Chicago

Andras Gilyen | Caltech

David Gross | University of Cologne

Michal Horodecki | University of Gdańsk

Stephen Jordan | Microsoft Research

Francois Le Gall | Nagoya University

Peter Love | Tufts University

Serge Massar | Université libre de Bruxelles

Nick Menicucci | RMIT University Melbourne

Ashley Montanaro (co-chair) | Phasecraft &
University of Bristol

Daniel Nagaj | Slovak Academy of Sciences

Ion Nechita | University of Toulouse

Michal Oszmaniec | Polish Academy of Sciences

Anna Pappa | Technical University of Berlin

David Perez-Garcia | Complutense University of
Madrid

Ben Reichardt | University of Southern California

Or Sattath | Ben-Gurion University

Peter Shor | Massachusetts Institute of
Technology

Graeme Smith | JILA & University of Colorado
Boulder

Kristan Temme | IBM Research

Barbara Terhal (chair) | Delft University of
Technology

Marco Tomamichel | National University of
Singapore

Michael Walter | University of Amsterdam

Chunhao Wang | Pennsylvania State University

Theodore Yoder | IBM Research

Beni Yoshida | Perimeter Institute

Henry Yuen | University of Toronto

Chosen for expertise and covering all topics (algorithms, cryptography, foundations, error correction, Shannon theory, gravity, many-body physics, NISQ, machine learning, complexity, etc.)

Review Process

- EasyChair using separate poster and talk tracks (professional license)
- Conflict of interest. Vice-chair (Montanaro) handles COIs of chair.
- **462** talk submissions (293, 349, 296 last QIPs).
- **Two reviews per paper** so 27-30 review per PC member.

General Statistics

Submissions	462
Accepted	111
Acceptance rate	0.24
Reviews	963
External reviewers	271
External reviews	338

Phases

1. Bidding and assignment
2. Reviewing (“give at most 5 papers a strong accept to borderline score”)
3. Alignment and Discussion (widely-varying scores, additional review cross-comparisons, coverage of topics, fairness/depth of reviews).
End: 60 talks accepted and 273 papers rejected.
4. Championing phase: select 15 papers among the remaining 121 papers. End: 44 additional talks with at least 4 champions accepted, so 104 in total.
5. Selection of short and long plenaries and best student paper prize using voting.

Reviewing

reviews for a paper	number of papers
2	425
3	35
4	2

More Statistics

- **104** PC-selected talks in total (71, 60, 57 in previous QIPs)
- 2 talks promoted to long plenaries
- 9 talks promoted to short plenaries
- 7 merged talks (of which one long plenary and 6 short contributed talks)
- 85 only-poster submissions (5 rejected as being off-topic/unclear on results)
- 100+ best-student paper award submissions (28 accepted, 1 selected)

Best Student Paper Award

Efficient classical simulation of random shallow 2D quantum circuits

John Napp, Rolando La Placa, Alexander Dalzell

(with Fernando Brandao and Aram Harrow)

Short Plenary Talk on Friday at 16.00 CET in Session 3 (Main Stage A)

Congratulations to the students!



Comments

Nov 20	Submission deadline; bidding begins on Nov 21
Nov 23	Bids due
Nov 24	Review assignments made
Dec 8	Recommended subreviewer deadline
Dec 13	Reviews due
Dec 14	Discussion begins
Dec 21–27	Xmas Break
Dec 28–31	Final Discussion
Jan 4	Authors notified

- The review process is **under stress** due to the fact that number of submissions is growing and review process involves (35) very busy scientists who have to allocate substantial time to reviewing in a relatively short period of time.
- Hard choices are made what to accept: ‘tip of the iceberg’. I and all PC members have worked hard to make the process as fair and ‘objective’ as possible to the extent possible.
- Accepting more talks would not necessarily lead to a better conference (**24%** versus 28%, 18% and 19% in recent years)
- [Personal] There often seems to be consolidation/priority towards certain types of ‘hard-core QIP’ results (in which the computer science and math aspect is important). This is fine, but we don’t just want to be a ‘quantum FOCS/STOC’. Always keep an eye on topic diversity and remember we don’t *have* a quantum computer.

Excerpts from the QIP charter

The role of the Program Committee (PC) is to select the best submitted papers and to put together a programme for the next QIP. The PC chair chooses the members of the PC (helped by advice from the SC), typically 15 or more people representing the broad range of subfields, including both computer science and physics. The PC chair determines (in cooperation with the SC) the rules for submission and puts out a call for submissions. The selection of talks among the submitted papers is competitive, with typically between 40 and 60 accepted submissions; it is recommended to keep the number of acceptances minimal but the PC has flexibility to accommodate exceptional breadth and quality of submissions in a given year. The PC can designate a few of the best submissions as “plenary” talks (given by “plenary speakers”), which are assigned more time in the program than regular contributed talks (given by “regular speakers”). All regular contributed talks are given in time slots ranging from 20-40 minutes.

Poster submissions will generally be accepted unless they are off-topic or clearly wrong, to enable people to obtain funds for travel. At the discretion of the PC, best poster prizes can be awarded.

QIP features a prize for the best student submission. A submission is eligible for this prize if and only if the main author(s) is/are a student(s) at the time of the submission and will present the work at QIP. Furthermore, a significant portion of the work (at least 60%) must have been done by said student(s), including the majority of the key ideas. Eligibility can only be indicated at the time of submission. All authors are notified if their paper has been labelled as eligible for the student prize, and have 14 days following submission to voice any disagreements about the paper’s nomination to the PC chair. The PC chair is free to ask for any clarifications regarding the students’ contributions at any time.